



Mechanism-driven artificial intelligence for food cold-chain control and governance

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ABSTRACT

Background: Food cold chains are critical infrastructures for safeguarding perishable foods, yet prevailing management strategies remain largely rule-based, relying on sparse environmental monitoring, conservative safety margins and endpoint inspection. Such approaches insufficiently capture product-level micro-environment variability and deterioration kinetics, resulting in recurrent temperature abuse, avoidable food loss, excessive energy consumption and elevated greenhouse-gas emissions.

Scope and approach: This review advances a mechanism-driven perspective on artificial intelligence (AI) in food cold-chain systems. Rather than treating AI as a generic optimisation tool, we examine how near-product, multimodal sensing can be coupled with kinetics-informed and physics-aware models to infer remaining shelf life, safety margins and quality risk in real time. We synthesise evidence on how these mechanism-linked outputs inform dynamic process control, quality-based logistics allocation, first-expired-first-out implementation and energy-aware operation. Particular emphasis is placed on model interpretability, uncertainty quantification, external validation and clearly defined applicability domains as prerequisites for trustworthy deployment.

Key findings and conclusions: Across reported studies, AI demonstrates tangible value only when tightly anchored to measurable micro-environments and explicit deterioration mechanisms, and when embedded within governed decision loops aligned with formal food-safety management systems. We distil core design principles for mechanism-driven cold-chain intelligence: robust sensing and calibration; physics-informed feature construction; uncertainty-aware modelling; transparent validation across products and scenarios; and controlled transferability. Under these conditions, AI-enabled systems can transition cold chains from static compliance frameworks to dynamic, risk-based governance structures, with realistic potential to reduce avoidable losses, optimise energy demand and lower emissions while enhancing transparency and accountability in cold-chain performance.

1. Introduction

Food cold chains are critical infrastructure for maintaining the safety and quality of perishable foods from post-harvest to consumption, spanning storage, transport, retail and use. Their effectiveness is grounded in food science: temperature and related micro-environmental conditions govern deterioration pathways (Chen et al., 2022; Han et al., 2026). When control weakens, microbial growth and toxin formation

can accelerate, while lipid oxidation, proteolysis, protein denaturation and moisture migration progressively degrade flavour, colour, texture and nutritional quality (Hoffmann et al., 2025; Shi et al., 2024). These deterioration processes are strongly matrix dependent: in fresh produce, respiratory heat production and ethylene sensitivity shape gas exchange and senescence, whereas in dairy and seafood, endogenous enzymes and microbiota interact with water activity and pH to define quality trajectories (Cui et al., 2024; Huang et al., 2023). Packaging further

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modifies the micro-environment through headspace composition, permeability and vent geometry, thereby mediating heat and mass transfer within the pack (Sonawane et al., 2025). Together, these mechanisms justify a product- and micro-environment-centred view of cold-chain performance: what matters is not only cold-room or trailer air temperatures, but also the exposure history at the scale of commodities, packs and pallets that ultimately governs residual quality and safety.

The importance of food cold chains is therefore closely tied to loss prevention and resource efficiency. Global estimates suggest that around 14% of food is lost between post-harvest and retail, with perishable commodities contributing disproportionately (FAO, 2019). Modelling and empirical analyses further indicate that insufficient or poorly managed cooling accounts for a substantial share of avoidable losses, and that strengthening cold-chain coverage, particularly rapid pre-cooling, stable early storage, temperature-controlled transport and well-managed retail display, can prevent many of these losses in fruits and vegetables, meat, dairy and seafood (Han et al., 2021; Sarr et al., 2021). When executed effectively, these measures not only reduce waste but also avoid the embedded land, water and greenhouse-gas emissions associated with food that would otherwise be discarded (UNEP, 2024). Cold-chain management is therefore not a minor operational detail but one of the most direct and scalable interventions for improving food-system resilience and environmental performance, yet this potential is still not fully realised in practice.

Despite decades of technological progress, conventional cold-chain practice still suffers from structural mismatches between what is measured, what is controlled and what ultimately determines product outcomes. Operational hand-offs, pre-cooling, loading, cross-docking, last-mile delivery and retail display continue to create gaps between product exposure and room- or compartment-level readings (Göransson et al., 2018; Mercier et al., 2017; Ndraha et al., 2018). Fixed setpoints, sparse single-point loggers and static date codes are weak proxies for the heterogeneous micro-environments within pallets and packages, the variability of pre-cooling histories and cumulative handling events that drive deterioration; consequently, “colder for longer” remains a common but energy- and carbon-intensive safeguard (Lamberty & Kreyenschmidt, 2025; Niu, Wang, et al., 2024). This measurement gap is not simply one of sensor precision: single-point loggers record accessible air conditions rather than the product- and package-scale micro-environments, cumulative thermal history and heat- and mass-transfer processes that govern deterioration. Moreover, many quality and safety changes are latent state variables rather than directly measurable signals (Hoffmann et al., 2023; Wang et al., 2023). Thus, even accurate point measurements may poorly represent true product exposure, and models trained on such sparse proxies may struggle to infer these latent states robustly under non-stationary conditions (Duan et al., 2026; Han et al., 2026). Classical kinetic and semi-empirical shelf-life models remain valuable for design and baseline planning, but they are typically calibrated under stable conditions and can lose accuracy under fluctuating conditions with sparse instrumentation (Cui et al., 2023; Rashvand et al., 2025). In many low- and middle-income regions, limited capital, unreliable power and scarce technical support further restrict cold storage and refrigerated transport, increasing spoilage risk and reinforcing disparities in market access and nutrition (Jackson et al., 2025; SEforALL, 2023; Singha et al., 2025). Taken together, these shortcomings arise as much from gaps in system design, measurement and governance as from a lack of computational sophistication, and they define the conditions under which more advanced analytics, including artificial intelligence (AI), can add practical value.

In this setting, AI can support a shift from compartment-level monitoring to product-centred decision-making only when it is explicitly anchored to deterioration mechanisms and near-product evidence. In this review, mechanism-driven AI does not refer simply to using physical variables such as temperature as model inputs. Rather, it refers to AI systems in which mechanism knowledge, including deterioration kinetics, heat and mass transfer relationships, product-state dynamics,

and operational safety boundaries, is used to shape feature construction, state estimation, model structure, optimisation constraints, or downstream decision rules. Under this definition, mechanism-driven AI occupies a middle ground between purely physics-based models and purely data-driven models: it preserves the flexibility of AI for heterogeneous, non-linear and partially observed cold-chain conditions, while retaining explicit links to causal processes that govern quality and safety outcomes.

We organise AI-enabled cold-chain intelligence into three tightly coupled functions. Prediction translates heterogeneous exposure histories into decision-grade signals such as product status, remaining shelf life (RSL), safety margins and quality or spoilage risk, rather than relying on fixed temperature thresholds (Rashvand et al., 2025; Shi et al., 2024). Optimisation embeds these signals within constrained decision-making across refrigeration, freezing, thawing and logistics, so that operational actions, such as setpoints, schedules, routing and stock allocation, satisfy explicit food-safety and quality requirements while balancing cost, energy use and emissions (Ding et al., 2025; Park et al., 2023; Qiao et al., 2024). Integration connects sensing and modelling outputs to execution through anomaly detection, intervention triggering, dynamic shelf-life (DSL) updating, stock rotation and traceable documentation across organisations and stages (Duan et al., 2026; Zou et al., 2025). Together, these functions show how AI can shift control from proxy air readings to product-relevant states, while translating predictions into auditable operational actions.

Critically, these functions are reliable only when deployment conditions are treated as first-class design constraints. This includes trustworthy micro-environment measurement and metadata, explicit safety constraints and failure modes, rigorous calibration and validation under realistic disturbances and heterogeneity, and disciplined transfer across products, packages, facilities and seasons. It also requires clear arrangements for data sharing, accountability, auditability and cross-actor coordination, especially in resource-constrained cold chains where failures are common rather than exceptional (Balakrishnan et al., 2025; Zhang, Liu, et al., 2025). In this review, governance therefore refers not merely to monitoring or control in isolation, but to the accountable organisation of sensing, prediction, decision thresholds, intervention execution and traceable documentation under formal food-safety, quality and operational constraints. This perspective is essential because AI creates practical value only when its outputs can be trusted, acted upon and reviewed within real cold-chain workflows.

Building on this perspective, this review provides a food-science-led systems synthesis of AI for food cold-chain control and governance. Rather than treating AI as a generic tool for digital monitoring or optimisation, we focus on how mechanism knowledge is embedded into sensing, prediction, optimisation and intervention workflows, and on how these workflows should be evaluated by verifiable, decision-relevant intervention outcomes rather than by model novelty alone. In doing so, this review makes three main contributions. First, it clarifies the conceptual boundaries of mechanism-driven AI by distinguishing it from purely physics-based modelling and purely data-driven learning. Second, it organises current evidence across product, process and logistics levels to show how deterioration mechanisms, product-relevant micro-environments and operational constraints can be linked to AI-enabled prediction, optimisation and integration. Third, it evaluates how these mechanism-linked decisions influence safety assurance, quality preservation, food loss and waste, energy use and greenhouse-gas emissions, while identifying the data, modelling and governance conditions required for responsible deployment at scale. Fig. 1 presents the conceptual framework that anchors this review, linking the limitations of conventional cold-chain management to a mechanism-driven AI workflow based on near-product sensing, embedded mechanism knowledge, decision-grade outputs, governed actions and trustworthy deployment conditions. Section 2 examines AI applications that generate decision-ready prediction and optimisation across product, process and logistics levels. Section 3 focuses on sensing and data

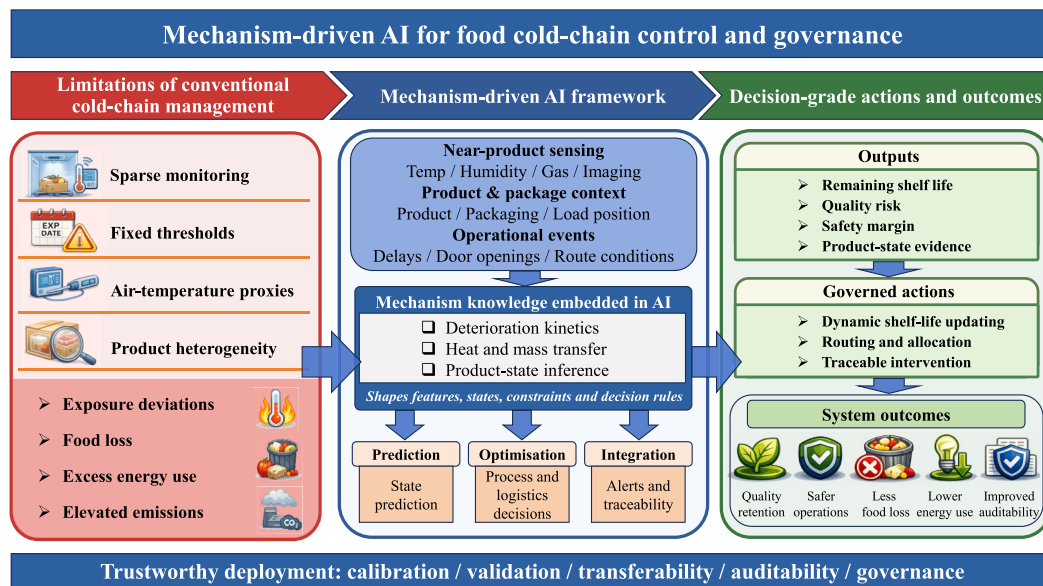


Fig. 1. Conceptual framework of mechanism-driven AI for food cold-chain control and governance. Conventional cold-chain management relies on sparse monitoring, static thresholds and air-temperature proxies, which inadequately capture heterogeneous product-level exposure and deterioration conditions. Mechanism-driven AI combines near-product sensing, contextual operational data and mechanism knowledge, including deterioration kinetics and transport-related state dynamics, to support prediction, optimisation and integration of decision-ready actions. When embedded in governed deployment conditions, these functions enable outputs such as remaining shelf life, quality risk and safety margin to be translated into traceable interventions and system-level benefits, including improved quality retention, reduced food loss and lower energy use.

infrastructure, anomaly detection and diagnostics, and closed-loop execution architectures for actionable interventions. Section 4 synthesises how AI-enabled control interacts with energy efficiency, decarbonisation and waste minimisation objectives. Section 5 discusses cross-cutting challenges and future directions for secure, verifiable and sustainable AI deployment.

2. AI in cold-chain process optimisation

AI creates the greatest value in food cold chains when it is anchored to measurable near-product micro-environments and explicit deterioration mechanisms, and when its outputs support decision-ready actions under real operational constraints. In this section, mechanism-driven AI refers not simply to the use of physical variables as inputs, but to prediction and optimisation workflows in which mechanism knowledge shapes inferred states, feature construction, optimisation constraints and downstream decision rules. We review these workflows at three levels: product, process and logistics. At these levels, AI is used to infer product status and RSL, optimise refrigeration, freezing and thawing trajectories, and support dynamic logistics decisions that account for quality loss, risk and shelf-life depletion. Section 2 therefore focuses on what is predicted or optimised, and how these outputs are linked to mechanism-based states, constraints and decision objectives. Section 3 then examines how these capabilities are operationalised through sensing, diagnostics and supervisory execution, while Table 1 summarises representative applications across Sections 2.1–2.3.

2.1. Predictive modelling of remaining shelf life and quality risk

At the product level, the core analytical task is to translate heterogeneous temperature-time exposure histories into forward-looking estimates of product state, remaining shelf life (RSL) and quality or spoilage risk that can directly support decisions. Classical shelf-life prediction has long relied on kinetic and semi-empirical models, for example Arrhenius-type temperature dependence combined with predictive microbiology or physicochemical deterioration models, and such approaches remain valuable because they are transparent, data efficient

and well suited to controlled or relatively homogeneous conditions (Cui et al., 2023; Gao et al., 2020). However, these models are usually calibrated under stable regimes and applied at batch level, so they can only coarsely represent heterogeneous trajectories, packaging-environment interactions and node-specific deviations in real cold chains, limiting item-, lot- or pallet-level updating under fluctuating and partially observed exposure conditions (Göransson et al., 2018; Mercier et al., 2017). Different quality attributes may also favour different modelling logics rather than a single universal approach. For example, in Sichuan sauerkraut, zero-order kinetics combined with the Arrhenius model better predicted physical indices, whereas a back-propagation artificial neural network better predicted chemical indices (Du et al., 2022). This comparison suggests that kinetic and data-driven models can play complementary rather than mutually exclusive roles in shelf-life prediction.

AI extends this toolbox most meaningfully when it does more than learn a non-linear mapping from temperature to outcome. In mechanism-driven prediction, the key issue is not simply whether physical variables are used as inputs, but whether mechanism knowledge helps define what should be reconstructed, predicted and acted upon. This can occur through virtual sensing of in-product temperature or other latent product states from sparse measurements, and through kinetics-informed or physiology-informed formulations that retain explicit links to deterioration processes. For example, a physics-informed neural network (PINN) has been used to predict postharvest avocado firmness by embedding a physiological framework for firmness loss within neural-network learning (van der Sman et al., 2026). A generic scientific machine learning (SciML) framework for fruit and vegetable quality prediction has also shown how domain knowledge can be integrated with neural networks to improve both prediction accuracy and temporal extrapolation under complex storage conditions (Guo et al., 2026). In this sense, mechanism-driven AI bridges purely physics-based modelling and purely black-box learning by retaining explicit links to deterioration processes while leveraging data-driven flexibility.

In real cold chains, predictive performance is often limited less by algorithm choice than by how faithfully product-relevant exposure and

Table 1

Representative AI applications in mechanism-driven cold-chain process optimisation across product, process, and logistics levels.

System layer	Decision context	Mechanism-relevant inputs/states	AI/optimisation approach	Decision output/action	Primary evaluation dimension	Representative result	Reference
Product level	Blueberry freshness assessment in cold-chain micro-environments	HACCP-derived O ₂ , CO ₂ , and C ₂ H ₄ signals linked to freshness deterioration	ML models (BP, RBF, SVM, ELM)	Freshness prediction	Prediction accuracy	90.87–94.01% accuracy, exceeding Arrhenius (85.10%)	Huang et al. (2023)
	Real-time RSL estimation for multiple marine fish species	Species, temperature, TVC, TVB-N, K-value, sensory scores, and E-nose/GC-MS features	ML models (BP, GA-BP, RBF, ELM) with platform deployment	Real-time RSL prediction	Prediction accuracy	Absolute errors <0.5; platform predicts RSL for five fish species in <1 s	Cui et al. (2024)
	Product-temperature estimation during multi-temperature urban delivery	Ambient temperature, initial food temperature, door status, and loading/unloading time	Improved ANN for virtual sensing	Real-time product-temperature estimation	State observability	Test RMSE 0.53–0.54 °C (chilled) and 0.61 °C (frozen); ~77–79% error reduction versus ambient-only inputs	Zou et al. (2023)
	DSL updating and quality-loss control in fruit supply chains	Pallet sensing, in-box temperature states, and kinetic shelf-life parameters	DT + AI (temperature prediction + kinetic shelf-life model)	DSL estimation and loss-control support	Shelf-life decision support	In-box temperature error < ±0.3 °C (2σ); shelf-life error < ±1.2 days; food loss reduced by 8.6–15.5% versus static shelf life XGBoost: MAE 0.0255 K, uncertainty ±0.036 K; MultilayerRNN predicted a 24.73% cooling-time reduction when airflow increased from 1 to 2 m/s	Zou et al. (2025)
Process level	Forced-air cooling of apples under varying package and airflow conditions	Cold-air temperature, air velocity, package/vent design, vent area, and apple initial temperature	ML and ANN models (e.g., XGBoost, SVR, MultilayerRNN)	Product-temperature trajectory prediction	Trajectory prediction accuracy	Energy model: MAE 0.0255 K, uncertainty ±0.036 K; MultilayerRNN predicted a 24.73% cooling-time reduction when airflow increased from 1 to 2 m/s	Kumar et al. (2025)
	Day-ahead setpoint scheduling for front-warehouse cold storage	Return-air temperature, compressor/air-cooler status, defrosting, outdoor conditions, and temporal features	LSTM prediction + PSO optimisation	Hourly temperature setpoint trajectory	Control effectiveness	Energy model: R ² = 0.93, MAE = 1.11 kWh; temperature model: R ² = 0.86, MAE = 0.072 °C; daily energy use reduced by 9.28% on average	Liu, Han, et al. (2025)
	Joint optimisation of colour protection and magnetic-field-assisted freezing for mushrooms	Formulation variables and freezing conditions linked to browning and ice-crystal damage	ANN-GA + magnetic-field-assisted freezing	Optimal protectant formulation and freezing condition	Quality retention	Optimal formulation: 0.17% L-cys, 0.07% VC, 1.13% CA; preferred field strength: 6 mT; drip loss reduced to 6.62% and ΔE to 9.96	Zhang et al. (2024)
	Non-destructive prediction of beef thawing degree and endpoint	Ultrasonic signal/velocity features calibrated against thermal-image-based thawing quantification	ML models (KNN, ANN, Extra-Trees, LightGBM)	Thawing degree estimation and endpoint detection	Endpoint identification	R ² > 0.9, up to 0.986; supports real-time non-destructive endpoint identification	Chen et al. (2024)
	Optimisation of SMF-assisted high-humidity low-temperature thawing for beef and lamb	Thawing temperature, relative humidity, and static magnetic field strength	BPNN + GA	Optimal thawing parameters	Quality retention	Thawing-loss prediction with MSE <0.1 and R close to 1; thawing efficiency improved by ~46.6% (beef) and ~32.5% (lamb)	Qiao et al. (2025)
Logistics level	Low-carbon heterogeneous multi-compartment cold-chain routing	Road-network data, fleet states, compartment capacities, and cost/emission-related vehicle attributes	DRL (DVS-AM)	Routing policy	Routing efficiency	Routing cost reduced by up to 6.32% and computation up to 1637× faster than a recent ant colony approach	Hu and Wang (2025)
	IoT-enabled multi-temperature delivery planning with incident-driven rerouting	IoT-detected incidents, delivery constraints, service windows, and product cooling-time windows	2PMGAO + fuzzy logic in IoT-MTDPS	Initial routing and dynamic rerouting	Real-time responsiveness	Travel-distance-related performance improved by 13%–27% versus traditional approaches; supports incident-aware rerouting	Tsang et al. (2021)

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Table 1 (continued)

System layer	Decision context	Mechanism-relevant inputs/states	AI/optimisation approach	Decision output/action	Primary evaluation dimension	Representative result	Reference
	Real environment-aware multi-objective cold-chain scheduling	Order, service, and real-captured transportation data, including actual distance, duration, and speed	MPMOACS	Scheduling and routing plan	Scheduling realism	Improved decision realism by incorporating real travel data; achieved the best overall performance against six recent multiobjective approaches	Wu et al. (2022)
	Dynamic cold-chain route re-optimisation under replenishment demand	Dynamic demand, replenishment states, time-window penalties, energy use, and product-quality-loss signals	Actor-Critic + RL-ALNS	Real-time route re-optimisation	Real-time responsiveness	Cost reduced from 2920.12 RMB to 2246.32 RMB; customer satisfaction increased from 0.8613 to 0.9346; solution time reduced from 28 s to 7.04 s	<link id=bib.zhu_et_al_2025b>Zhu, Wang, et al., 2025; </link>
	Sustainable cold-chain intelligent green scheduling	Operational signals related to cost, vehicle use, carbon emissions, traffic restrictions, and multi-level time windows	IGSS-CCL with Dijkstra + NSGA-III	Dispatching and scheduling decisions	Sustainability performance	Supports joint optimisation of operating cost, vehicle use, and carbon emissions; multi-objective planning outperformed corresponding single-objective settings	Shi et al. (2022)

Abbreviations: AI, artificial intelligence; RSL, remaining shelf life; DSL, dynamic shelf-life; DT, digital twin; ML, machine learning; ANN, artificial neural network; BPNN, back-propagation neural network; BP, back propagation; RBF, radial basis function; ELM, extreme learning machine; SVM, support vector machine; TVC, total viable count; TVB-N, total volatile basic nitrogen; GC-MS, gas chromatography-mass spectrometry; FAC, forced-air cooling; PSO, particle swarm optimisation; GA, genetic algorithm; KNN, k-nearest neighbours; SMF, static magnetic field; DRL, deep reinforcement learning; DVS-AM, dynamic vehicle state attention model; 2PMGAO, two-phase multi-objective genetic algorithm optimiser; IoT-MTDPs, Internet of Things-based multi-temperature delivery planning system; MPMOACS, multiple population-based multiobjective ant colony system; RL-ALNS, reinforcement learning-guided adaptive large neighborhood search; IGSS-CCL, intelligent green scheduling system for cold chain logistics; NSGA-III, non-dominated sorting genetic algorithm III.

latent deterioration states are represented. For this reason, mechanism-driven prediction increasingly relies on multimodal sensing and virtual sensing to reconstruct states that are more closely linked to deterioration than accessible air-point readings alone. A growing body of work has therefore moved beyond temperature-only inputs towards packaging descriptors, imaging and spectroscopy signals, gas variables and other non-destructive indicators, while also recognising persistent challenges in data availability, deployment cost and generalisability ([Rashvand et al., 2025](#)). Representative applications illustrate that the value of additional information lies not simply in increasing input dimensionality, but in better capturing the underlying product state. For stored garlic, machine-learning models identified temperature as a dominant driver while also revealing moderating packaging effects ([El-Mesery et al., 2025](#)); for blueberries, combining sensor streams with image-derived features improved freshness prediction over temperature-only baselines ([Huang et al., 2023](#)); and for seafood logistics, integrating microbial, chemical and sensory indicators has supported near day-scale shelf-life prediction within operational platforms ([Cui et al., 2024](#)). Importantly, ambient-air records may diverge from in-product temperatures during multi-temperature distribution, and virtual sensing has therefore been used to reconstruct product-relevant thermal histories as a prerequisite for reliable RSL estimation rather than as a separate engineering detail ([Zou et al., 2023](#)).

Model selection should therefore be guided not only by data structure, but also by the type of product state being inferred and the decision context in which predictions will be used. For small but well-curated datasets with engineered features, ensemble learners such as random forests and gradient boosting remain practical tools, whereas sequence models, especially long short-term memory (LSTM) and gated recurrent unit (GRU) networks, are better suited to long and irregular exposure histories ([Shi et al., 2024](#)). However, no model family is mechanism-driven by default. Deployment-grade prediction requires

that state definitions, targets and validation protocols remain aligned with deterioration logic and operational use, together with external validation across batches, seasons, suppliers and facilities and calibrated uncertainty for risk-aware decisions ([Balakrishnan et al., 2025](#)). Uncertainty-aware dynamic shelf-life (DSL) frameworks further extend this logic by treating RSL as a stochastic process under real cold-storage conditions ([Chen et al., 2023](#)).

Crucially, predictive modelling becomes operationally meaningful only when its outputs can update actionable product-state estimates and support downstream allocation, labelling and inventory decisions. Rather than relying on fixed expiry dates and binary pass/fail rules, mechanism-linked models can generate RSL curves, risk trajectories or uncertainty-aware thresholds at lot or pallet level. DSL concepts translate these outputs into adaptive decision rules, enabling quality-based allocation, first-expired-first-out (FEFO)-style stock rotation and more targeted waste-reduction strategies ([Chen et al., 2023](#); [Mercier et al., 2017](#); [Wu, Zou, Chen, et al., 2025](#)). Hybrid approaches, including digital-twin coordination combined with data-driven in-box temperature prediction and kinetic shelf-life estimation, further show that the practical value of mechanism-driven prediction lies in linking product-state estimation directly to intervention design ([Zou et al., 2025](#)).

2.2. Optimisation of refrigeration, freezing and thawing processes

At the process level, refrigeration, including pre-cooling, as well as freezing and thawing, shape the thermal and physicochemical histories that determine downstream quality retention, microbiological safety and remaining shelf life. These operations influence cooling uniformity, phase-change progress, water redistribution, tissue damage, drip loss, texture and colour stability, thereby defining a process-level “quality budget” that subsequent logistics can either preserve or further erode

(Alabi et al., 2020; Li et al., 2018). Despite their central role, these steps are still widely managed using recipe-like routines, including fixed room setpoints and conservative pre-cooling times, standard freezing programmes, and time- or ambient-based thawing schemes (Jiang et al., 2023). Such routines are simple and auditable, but they share several structural weaknesses. First, they often treat air or equipment conditions as proxies for product states, even though room temperature may not represent product core temperature during cooling, a single freezing residence time may not reflect phase-change progress across different sizes, packaging formats or loading patterns, and thawing time may fail to capture internal temperature uniformity. Second, they respond poorly to variation in initial thermal history, product geometry, airflow distribution and evolving equipment performance, which commonly produces intra-batch heterogeneity, with hotspots and over-processed zones coexisting (Schemminger et al., 2025; Verreydt et al., 2024). Third, because process-relevant quality endpoints are difficult to quantify online, operations are often tuned around throughput and conservative safety margins, while quality losses become visible only after the process has ended (Jiang et al., 2023).

Mechanism-driven AI helps address these limitations by reframing refrigeration, freezing and thawing as constrained decision problems centred on product-relevant states and quality-governed process trajectories. The first recurring building block is state estimation, often implemented through virtual sensing. Here, learning models infer states that are not directly observable online but are mechanistically relevant to process outcomes, such as product core temperature, cooling progress, phase-change status, thawing degree and spatial or temporal uniformity, from sparse measurements and operational signals (Duan et al., 2026). These inferred states matter because they are more closely linked than room-level readings to heat transfer, temperature equalisation, phase transition and the quality deterioration pathways that unfold during processing. The second building block is surrogate modelling combined with constrained optimisation. In this setting, supervised models map controllable variables, such as setpoints, staged temperature profiles, airflow or humidity settings, and auxiliary-field parameters, to multiple process outcomes, while an optimiser identifies implementable policies, including dynamic setpoint trajectories, staged profiles and stop criteria, under equipment, throughput and safety constraints (Qiao et al., 2024). Energy or operating cost may be incorporated as secondary objectives or constraints, but the defining contribution in this section is that AI makes quality-centred tuning of process trajectories feasible at industrial timescales and under realistic operational conditions.

Recent studies illustrate how this logic operates differently across refrigeration, freezing and thawing. In refrigeration and pre-cooling, the central challenge is often the gap between measured air conditions and actual product cooling histories. Machine-learning models have therefore been used to predict product temperatures during forced-air cooling, enabling batch-specific adjustment of cooling duration and airflow management without dense instrumentation (Kumar et al., 2025). Once such predictive capability is available, it can be embedded in constrained control frameworks to generate feasible trajectories rather than relying on static setpoints. For example, an LSTM framework coupled with particle swarm optimisation and trained on real cold-room data was used to optimise hourly setpoint trajectories, improving thermal stability and thereby supporting better product preservation under operational disturbances (Liu, Chen, et al., 2025). In freezing, the dominant challenge is that quality depends not only on target temperature, but also on the dynamics of phase transition and the resulting structural damage. AI has therefore been used to learn non-linear relationships between freezing conditions and multiple post-process quality attributes, and then search for improved operating conditions or freezing curves. An artificial neural network (ANN) coupled with a genetic algorithm (GA) exemplifies this strategy by identifying quality-oriented freezing settings that balance competing outcomes such as texture retention, drip loss and colour change (Zhang et al., 2024). In

thawing, by contrast, major failure modes arise from endpoint uncertainty and internal non-uniformity, because surface warming does not necessarily indicate that the interior has reached a suitable and safe thawed state. Accordingly, ML combined with ultrasonic measurements has been used to predict thawing degree and endpoints to support adaptive termination or staged control (Chen et al., 2024), while back-propagation neural network (BPNN) and GA frameworks have further been applied to optimise composite thawing parameters for improved efficiency and post-thaw quality (Qiao et al., 2025).

Overall, AI shifts refrigeration, freezing and thawing from static, rule-based routines towards mechanism-informed, product-relevant and quality-governed process control. Its value does not lie simply in replacing conventional settings with more complex algorithms, but in enabling auditable optimisation of trajectories and stop criteria against multiple quality targets under practical constraints. In this way, process operations become better aligned with the product states that matter for subsequent shelf-life, risk and logistics decisions, while models can be progressively improved as sensing, validation and operational data accumulate.

2.3. Quality-aware real-time adjustments in cold-chain logistics

At the logistics level, cold-chain decisions govern whether the remaining quality budget inherited from prior processing is preserved, redistributed or rapidly depleted during distribution. Accordingly, cold-chain performance must be evaluated by product condition at delivery, rather than by on-time metrics alone, because cumulative exposure, handling events such as door openings, and node-level delays can degrade quality even when service windows are met (Göransson et al., 2018; Wu et al., 2022). These effects are amplified in cross-docking, retail backrooms and last-mile distribution, where dwell-time variability and traffic disturbances increase temperature excursions, cumulative exposure uncertainty and heterogeneity in product condition across loads and stops (Göransson et al., 2018). As a result, quality outcomes may diverge substantially from static date codes, fixed dispatch rules and route plans that optimise distance or time alone, motivating logistics control that explicitly accounts for deterioration risk, remaining shelf life and operational trade-offs rather than treating quality as an end-point check (Shi et al., 2022; Stellingwerf et al., 2021).

Quality-aware, real-time logistics control therefore requires continuous reprioritisation as product condition and operating context evolve. Reinforcement learning (RL) and related online optimisation methods provide one coherent framework for this purpose by treating routing, dispatch and reassignment as sequential decisions under uncertainty. In typical formulations, the state encodes vehicle locations, remaining capacities, time windows and available slack, and load composition, including multi-temperature compartments, together with real-time traffic and environmental signals and product-relevant variables supplied by prediction modules, such as shipment-level condition estimates, RSL trajectories or quality-loss proxies (Hu & Wang, 2025; Tsang et al., 2021). Actions may include resequencing stops, dynamic rerouting, order reassignment across vehicles, depots and compartments, dispatch reprioritisation, and the triggering of mitigations such as alerts, expedited handling or conservative fallback policies when predicted risk increases (Ahmadi-Javid et al., 2023; Tsang et al., 2021). Rewards can jointly penalise tardiness and logistics cost while explicitly penalising predicted quality loss, shelf-life depletion or non-compliance risk, with energy and emissions terms included as secondary objectives or constraints when needed (Shi et al., 2022; Stellingwerf et al., 2021). Because cold chains are safety critical, compliance must be enforced through constrained action spaces and explicit feasibility constraints, including temperature limits, compartment compatibility and service windows, so that real-time adjustment does not generate unsafe or operationally infeasible trajectories (Ahmadi-Javid et al., 2023; Hu & Wang, 2025).

Empirical support for this direction is strongest when both objective formulation and the monitoring-to-action loop are explicitly tied to

product deterioration rather than generic logistics efficiency. For objective design, the quality-driven vehicle routing problem embeds time- and temperature-dependent kinetic decay, based on Arrhenius-type deterioration, into route optimisation. In a cooperative retail cold-chain case, it shows that multi-stop routes can shift disproportionate quality loss towards late-route customers even when distance- or cost-based objectives appear acceptable, thereby providing a direct template for reward shaping and route evaluation around deterioration risk rather than transport efficiency alone (Stellingwerf et al., 2021). For the monitoring-to-action loop, a connected multi-temperature delivery planning system couples real-time location and environment monitoring with incident-triggered rerouting under multi-temperature requirements, illustrating how updated traffic and environmental states can be translated into dispatch updates and condition-aware replanning in perishable e-commerce logistics (Tsang et al., 2021). At a larger scale, real environment-aware scheduling with multisource data association further shows how heterogeneous signals can be fused into actionable scheduling and reassignment decisions under realistic operating conditions (Wu et al., 2022).

Within this broader decision framework, learning-based policies, including deep RL, are beginning to support cold-chain routing and dispatch under increasingly realistic constraints. A deep RL route-optimisation model for multi-compartment cold-chain distribution demonstrates that learned policies can generate routing decisions under compartment constraints and distribution complexity (Hu & Wang, 2025). For agricultural products under replenishment demand, an RL model for dynamic distribution route optimisation provides evidence that learning-based decisions can adapt routes and assignments in

response to evolving order and demand states (Zhu, Liu, et al., 2025). In practice, however, safe deployment typically proceeds through offline learning and simulation-based stress testing, with restricted action bounds, fallbacks and human oversight, because unsafe exploration on real loads is unacceptable (Shi et al., 2022; Wu et al., 2022). Evaluation should therefore extend beyond cost and on-time performance to include expected quality retention, such as RSL depletion, time outside target temperature ranges, food loss risk, intervention effectiveness, and associated energy and emission outcomes (Shi et al., 2022; Stellingwerf et al., 2021). The sensing, connectivity and closed-loop execution architectures needed to operationalise these monitoring-to-decision pipelines are discussed in Section 3.

3. AI in real-time cold-chain sensing and control

AI creates operational value in food cold chains only when predictions and optimisation outputs can be deployed through reliable sensing, contextual diagnostics, and constrained execution. Building on Section 2, this section therefore shifts from what AI predicts or optimises to how those capabilities are operationalised in real cold-chain systems. It examines three linked elements of a sensing-to-execution loop: sensing and data infrastructure for decision-grade observability (Section 3.1), anomaly detection and predictive maintenance for resilient and safe intervention gating (Section 3.2), and supervisory control for cold storage and transport, where validated insights are converted into auditable actions under explicit safety and feasibility constraints (Section 3.3). Fig. 2 illustrates this operational sensing-to-execution loop and clarifies the entry point of mechanism-driven AI in Section 3:

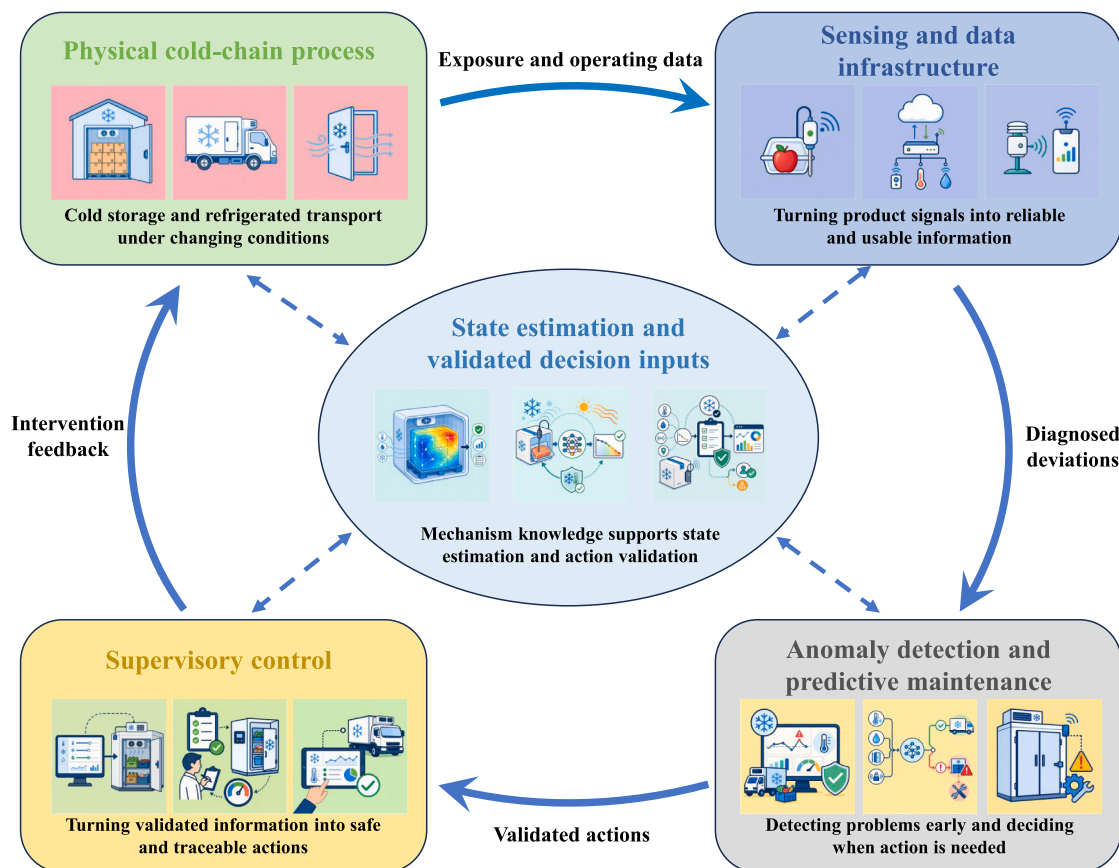


Fig. 2. Operational sensing-to-execution loop for mechanism-driven AI in real-time food cold-chain control. Exposure and operating data from the physical cold-chain process are converted into reliable information through sensing and data infrastructure, interpreted through anomaly detection and predictive maintenance, and translated into validated actions by supervisory control. The central hub highlights the mechanism-driven role of state estimation and action validation, linking near-product signals, deterioration-relevant states and operational constraints to safe, traceable interventions.

mechanism knowledge does not operate as a separate modelling block, but supports state estimation and action validation by linking near-product sensing, deterioration-relevant states and operational constraints.

3.1. Sensing and data infrastructure for End-to-End visibility and decision-grade observability

Cold-chain systems operate under tight thermal and time constraints, where small, unobserved deviations can rapidly translate into quality and safety losses (Han et al., 2026). Yet conventional monitoring often relies on a few compartment air probes, while products experience highly localised micro-environments shaped by pallet and vehicle stratification, door-opening transients and packaging heterogeneity (Elashmawy et al., 2025; Lamberty & Kreyenschmidt, 2025; Verreydt et al., 2024; Zou et al., 2023). Because deterioration is driven by local heat and mass transfer, respiration and condensation, air-point logs can be weak proxies for true exposure histories (Hoffmann et al., 2023; Wang et al., 2023). Therefore, end-to-end visibility should be framed as decision-grade observability: capturing product- and package-representative signals and exposure trajectories that are time-aligned, calibrated, quality-assured, traceable, and useable for downstream prediction, diagnostics and execution, rather than simply increasing data volume.

From a mechanism-driven perspective, the key sensing question is not only what can be measured directly, but also which states should be inferred because they are more tightly linked to deterioration than accessible air readings alone. To close the gap between air-point logs and true product exposure, sensing for visibility should prioritise product and package-scale micro-environment dynamics that are tightly coupled to deterioration mechanisms. Two temperature-focused studies illustrate how near-product measurements reveal exposures that compartment probes miss: a miniaturised Peltier element quantified bidirectional fruit–environment heat flux during postharvest storage even when fruit–air temperature differences were <0.5 °C, capturing short-lived thermal stresses (Hoffmann et al., 2023), while conformal surface sensing exposed multi-hour lags between ambient and package air and true fruit temperature and highlighted within-load non-uniformity (Wang et al., 2023). Complementary gas sensing extends visibility beyond thermal stress to chemical freshness and spoilage in sealed cold environments: a Pt@Fe–TiN fuel-cell H_2S sensor remained operational at -18 °C with ppb-level detection for continuous meat spoilage monitoring (Zhu, Liu, et al., 2025), and an “artificial olfaction” chip using a 3×3 gas-sensor array discriminated multiple spoilage-related volatiles (e.g., ammonia, ethanol, trimethylamine and hydrogen sulfide) to enable real-time freshness evaluation under refrigeration (Tang et al., 2026). Together, these examples shift monitoring from compartment-level compliance logging towards product-relevant state indicators that can be reliably used as model inputs and control-relevant signals, or used to reconstruct latent states that better represent deterioration-relevant exposure histories.

Turning these product- and package-representative signals into end-to-end visibility is primarily a data-infrastructure problem: heterogeneous streams must be captured with context, aligned in time and made auditable before they can support prediction, diagnostics or control (Han et al., 2026; Jia et al., 2025). In practice, deployments adopt a layered edge–cloud pipeline in which endpoints timestamp and tag measurements, edge gateways buffer data under intermittent connectivity and perform light pre-processing (e.g., resampling and basic integrity checks), and cloud services integrate streams for storage, monitoring and model execution. A representative implementation couples streaming ingestion with online prediction: sensor feeds are forwarded to a Kafka-based pipeline where services subscribe to perform calibration and resampling and execute condensation and quality-loss models; predicted outputs are then persisted to a time-series store for real-time visualisation, and the resulting predictions are reported to

track trends observed in storage trials (Sonawane et al., 2025). At the platform level, Internet of Everything (IoE)–digital twin service architectures formalise this pipeline into reusable layers, sensing, gateway and edge computing, cloud data and analytics, and application services, so that visibility can be scaled across sites and use cases rather than built as one-off systems (Wu et al., 2023). For cross-organisation visibility, infrastructure must also embed governance and connectivity choices; supervisory frameworks highlight low-power wide-area communications and privacy-preserving mechanisms as prerequisites for reliable monitoring and trusted data sharing (Yu, Zhang, et al., 2025). These foundations reduce false alarms and missed events in anomaly detection (Section 3.2) and enable supervisory control to act on constraint-consistent, auditable inputs rather than noisy proxies (Section 3.3), thereby turning sensing infrastructure from a monitoring layer into the operational entry point for state-aware diagnostics and constrained execution.

3.2. Anomaly detection and predictive maintenance for operational resilience and safe control

Operational resilience in the cold chain depends on detecting deviations early and interpreting them in context, because the same temperature excursion can be benign during loading yet critical when it persists off-setpoint or recurs on specific routes (Loisel et al., 2021; Ndraha et al., 2018). With dense streams from Internet of Things (IoT) devices (temperature/relative humidity (RH), door events, equipment signals, routes and loads), anomalies are inherently contextual: short spikes during hand-offs may be acceptable, whereas persistent off-setpoint exposure or repeated setpoint failures can indicate high risk (Gillespie et al., 2023). Because fault events are rare and weakly labelled, deployments often start with scalable stream detectors (windowed statistics plus learning-based scores) to flag suspect episodes and progressively accumulate labels for targeted supervised models (Martínez-Heredia & Ventura, 2025). Crucially, these detections must be calibrated to gate downstream interventions, so that control actions are triggered when risk is credible and constraint-relevant, rather than by noisy transients, and so that deviations are interpreted according to their likely consequences for product condition, control feasibility and operational resilience rather than by threshold exceedance alone.

For last-mile distribution, a cost-effective ANN-based over-temperature alarm service showed that high accuracy can be achieved with a compact feature set while explicitly trading off detection performance against sensing and communication costs (Meng et al., 2024). At the device and edge level, multi-threshold architectures that combine sensor readout, on-device processing (e.g., analogue-to-digital conversion and microcontroller logic) and wireless communication can provide graded “normal–pre-alert–alert” warnings with closed-loop actuation, improving responsiveness under field conditions (Zheng et al., 2026). At the supply-chain scale, data-driven virtual sensor systems (e.g., bidirectional long short-term memory (BiLSTM) with attention) can fuse multisource temperature, operational and attribute data to estimate hard-to-instrument temperature states and trigger alerts by comparing predicted values against thresholds, thereby extending monitoring coverage beyond sparse physical sensors (Duan et al., 2026). Fig. 3 summarises representative solutions spanning service-level alarms, device-level thresholded warning modules and supply-chain-level virtual sensing for resilient operations, showing that anomaly detection becomes more decision-relevant when alerts are linked to inferred product-relevant states rather than raw excursions alone.

Anomaly detection also underpins fault detection and diagnosis (FDD) and predictive maintenance for refrigeration assets in supermarkets, cold stores and vehicles. AI-assisted FDD has been used to detect incipient refrigerant leakage, icing and frosting, heat-exchanger fouling, valve malfunctions and sensor bias, reducing energy waste and the likelihood of temperature drift at product locations (Fallmann, Lösch, et al., 2023; Soltani et al., 2022). A large-scale deployment

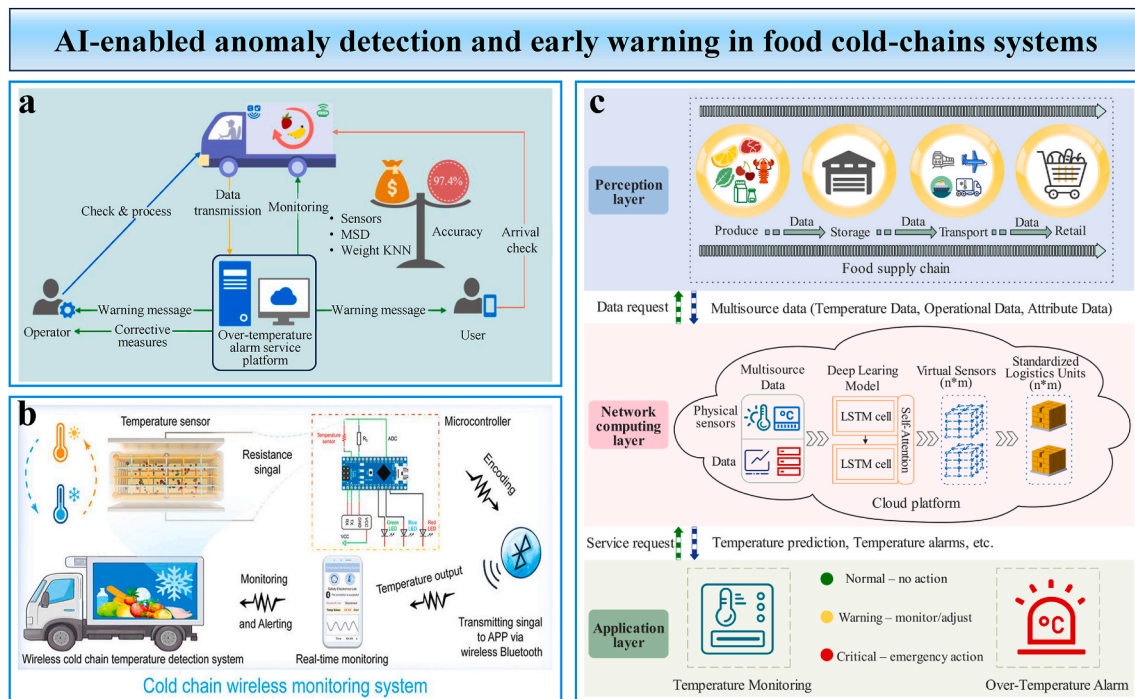


Fig. 3. AI-enabled anomaly detection and early warning in food cold-chain systems. (a) A cost-effective over-temperature alarm service platform for refrigerated transport, showing data acquisition, transmission, monitoring, and warning-response loops (Meng et al., 2024). (b) An integrated monitoring architecture combining flexible sensing, wireless communication, and multi-threshold logic to provide graded normal-pre-alert-alert warnings and on-site notifications (Zheng et al., 2026). (c) A virtual sensing framework across produce, storage, transport, and retail, where multisource data are fused by deep sequence models to infer temperatures at sensor-inaccessible locations and support prediction and threshold-based alarms (Duan et al., 2026).

across 100 commercial refrigeration systems reported >95% diagnostic accuracy for most fault types and ~17% average energy savings when AI-assisted FDD was integrated with monitoring-based commissioning (Lee et al., 2022). Typical workflows extract features from multi-sensor telemetry (pressures, temperatures, currents and door patterns), infer fault type and severity and, where feasible, remaining useful life (RUL), and then prioritise interventions by combined impacts on safety, quality preservation and energy performance rather than fixed service intervals (Zhu et al., 2024). By translating fault signatures into severity- and RUL-informed priorities, predictive maintenance becomes a control-relevant lever that stabilises operating conditions and reduces recurrent excursions, while also preserving the sensing quality, equipment reliability and operating conditions required for trustworthy supervisory execution. In this sense, predictive maintenance is not only an asset-management function, but also a mechanism-relevant safeguard for maintaining the validity of downstream control decisions.

3.3. Supervisory control for cold storage and transport

Low-level temperature regulation in cold storage facilities and refrigerated vehicles remains dominated by fixed setpoints with on-off logic or proportional-integral (PI)/proportional-integral-derivative (PID) loops, a pragmatic choice that prioritises robustness but often treats the load as thermally uniform (Niu, Liu, et al., 2024; Park et al., 2023). In reality, exposure is spatially and temporally heterogeneous: airflow paths, loading patterns and door-opening transients create within-compartment gradients that sparse air probes may not represent, while product temperatures respond with location-dependent lags (Duan et al., 2026; Hoffmann et al., 2023). Full-scale monitoring in commercial citrus operations further shows that both pre-cooling and refrigerated trailer transport can exhibit pronounced air–fruit heterogeneity and shipment-to-shipment variability, including hotspot formation that is not inferable from a single rear-trailer measurement (Verreydt et al., 2024). These observations motivate supervisory control

not as “making the setpoint smarter” in isolation, but as an execution layer that converts visibility into safe, interpretable actions across assets and hand-offs. Accordingly, unlike the optimisation tasks discussed in Section 2, the focus here is not on deriving control policies in the abstract, but on how observability from Section 3.1 and diagnostic gating from Section 3.2 are translated into validated, constrained and auditable supervisory actions under real operating conditions.

In practice, supervisory control complements fast local loops (on-off/PI/PID) with a slower decision layer that periodically adjusts higher-level variables, setpoint trajectories within allowable bands, airflow and evaporator modes, pull-down strategies and defrost scheduling, based on estimated thermal states, anticipated events (loading and door openings) and explicit safety constraints. Model predictive control (MPC) provides a natural backbone because it optimises over a horizon while enforcing hard bounds on temperature and equipment limits. For mobile refrigeration, a control-oriented low-order hybrid model that explicitly represents door openings as genuine changes in system dynamics offers reliable process insight for model-based control, soft sensing, and fault detection and isolation (Fallmann, Poks, & Kozek, 2023). Building on such representations, hybrid MPC can incorporate operational flexibilities (e.g., thermal inertia and temporary storage effects, permissible temperature bands and expected door events) while maintaining temperature restrictions under realistic last-mile disturbances (Fallmann, Lösch, et al., 2023). For stationary refrigeration assets, supervisory MPC can be implemented as “setpoint management”: adjusting cabinet and room setpoints within safe ranges rather than directly overriding embedded controllers, which is attractive for retrofits and safety-critical deployment (Mazare, Izadi-Zamanabadi, & Ramezani, 2026). In this sense, supervisory control should be understood less as a new optimisation problem than as a structured execution layer that filters, schedules and applies allowable actions while preserving the integrity of local control loops and operational safeguards.

AI methods strengthen supervisory control primarily by improving state estimation and robustness, not by relaxing constraints. Virtual

sensing can increase effective spatial observability when only sparse physical sensors are feasible; for example, a one-sensor-per-pallet virtual temperature sensor was developed to detect temperature breaks with minimal instrumentation, enabling actionable monitoring inputs for supervisory decisions (Mortazavi et al., 2025). Digital twins further support safe deployment by enabling offline training, what-if testing and parameter updating as equipment and operating conditions drift; recent work on cascade refrigeration systems emphasises real-time data-driven updating to maintain consistency between digital and physical twins (Li et al., 2024), while broader heating, ventilation, air conditioning and refrigeration (HVAC + R) digital-twin reviews synthesise how such twins integrate monitoring, diagnostics and optimisation in safety-constrained environments (Ibrahim et al., 2026). Overall, Section 3.3 closes the monitoring-to-action loop by specifying how decision-grade observability (Section 3.1) and fault-aware signals from anomaly detection and predictive maintenance (Section 3.2) translate into constrained, auditable supervisory decisions with traceable decision logging for both cold stores and refrigerated transport, including action validation, fallback logic and human review where needed, while energy-centric objectives are addressed in Section 4.1.

4. AI in enhancing cold-chain energy efficiency and sustainability

Sustainability in food cold chains is not an energy-saving problem alone, but a quality-, safety- and carbon-constrained decision problem. Cold-chain operation reduces food loss by slowing deterioration, yet it also consumes electricity, transport fuel and refrigerants, generating greenhouse-gas emissions. Excessively conservative cooling may increase energy and carbon burdens, whereas aggressive energy saving may accelerate quality loss and increase the embedded emissions of wasted food. Sustainable cold-chain control therefore requires coordinated optimisation of product quality, economic efficiency and environmental impact, rather than isolated minimisation of electricity use (Chen et al., 2022; Chen et al., 2026).

From a mechanism-driven AI perspective, the key role of intelligent control is to link near-product evidence, deterioration-related state estimates and operational constraints to sustainability-relevant actions. Real-time sensing, adaptive decision-making and closed-loop execution can help determine when cooling is necessary, how much thermal flexibility is safe, when faults require maintenance and how products should be allocated or rerouted before quality loss becomes unacceptable (Han et al., 2026). This chapter therefore examines three coupled pathways: smart energy management in cold-storage facilities, carbon-footprint reduction across cold-chain operations, and waste minimisation through quality-preserving decisions. Across these pathways, AI should be evaluated by verifiable intervention outcomes, including quality-safe energy savings, reduced emissions, lower food loss and auditable decision records.

4.1. Smart Energy management in cold-storage facilities

Cold-storage facilities are among the most electricity-intensive nodes in food cold-chain systems, largely because temperature protection is commonly delivered via conservative static setpoints, reactive defrost and limited exploitation of thermal inertia. Under fluctuating ambient conditions, door openings and time-varying electricity prices, this “safety-margin” operation can translate into avoidable overcooling, post-defrost power peaks and mismatched compressor and fan scheduling. Smart energy management should therefore be framed as a quality- and safety-constrained control problem: energy-saving actions are useful only when they maintain product-relevant temperature conditions, preserve thermal stability and avoid shifting risk from electricity consumption to quality loss.

Recent evidence shows that control strategy design itself can deliver substantial economic benefits even when overall cooling delivery is

similar. Comparing alternative cooling-supply control strategies in frozen and chilled rooms, Niu, Wang, et al. (2024) reported that electricity bills could drop to $\sim 0.35\times$ (frozen room) and $\sim 0.5\times$ (chilled room) of the comparator strategy under strong peak–valley pricing, highlighting the value of tariff-aware supervisory control. Prediction-based optimisation provides a more adaptive intermediate strategy. For example, an LSTM-based framework for daily energy and temperature prediction, coupled with particle swarm optimisation, reduced daily energy consumption by an average of 9.28% in an urban front-warehouse cold store while maintaining safe storage conditions through optimised hourly setpoints (Liu, Chen, et al., 2025). Moving beyond rule-based and prediction-optimisation strategies, deep reinforcement learning (DRL) has demonstrated large real-world savings when temperature constraints are explicitly enforced. In an actual cold-storage facility, a modified deep deterministic policy gradient (DDPG) controller achieved a 47.64% reduction in energy consumption relative to conventional PID control while maintaining the target temperature range (Park et al., 2023). In larger-scale settings, continuous-DDPG control further delivered $\sim 13.6\text{--}16\%$ energy-use reduction, with the best configuration reaching 19.8% savings; critically, results emphasised that sensor placement and the choice of input signals strongly affect achievable savings by improving responsiveness around defrost-related disturbances and reducing post-defrost peaks (Park et al., 2025). These studies indicate that smart energy management in cold stores is increasingly a joint design problem of control policy plus sensing strategy, rather than an algorithm-only upgrade.

Digital-twin and model-based optimisation offer a complementary pathway, especially for complex low-temperature systems where setpoints interact across stages. Li et al. (2025) developed a cascade refrigeration-system digital twin with bidirectional data flow (IoT plus cloud–edge computing) and reported an average coefficient of performance (COP) increase of 7.2% after integrated optimisation based on operating data, illustrating how twin-enabled optimisation can translate telemetry into actionable efficiency gains. In parallel, flexibility-enhancing hardware can amplify control benefits: AI has been used to support thermal energy storage design and operation (e.g., screening phase-change material properties and optimising charge–discharge strategies) so that storage buffers temperature excursions during door openings and mitigates grid-price peaks (Jin et al., 2026; Liu, Han, et al., 2025). Field-oriented retrofits also remain important: upgrades at the envelope and system levels (e.g., insulation, control modernisation and renewable integration) can reduce peak load and improve system efficiency, supporting the case for “AI + physical measures” rather than software-only interventions (Singha et al., 2025). Finally, energy management should be fault aware because operational abnormalities can silently erode efficiency. In a large cold-store refrigeration system, resolving defrost-related faults after automated state identification was associated with $\sim 18.3\%$ per day reductions in both energy consumption (~ 1800 kWh) and carbon emissions (~ 1494 kg) (Zhu et al., 2024), indicating that maintenance-informed optimisation can be as impactful as advanced scheduling in some sites.

These approaches differ substantially in suitability, data requirements and deployment risk. Rule-based and tariff-aware supervisory control is attractive for retrofits because it is interpretable, inexpensive and data-light, but it mainly exploits predictable price or load patterns and may respond poorly to unplanned disturbances. Prediction-optimisation frameworks provide a practical middle ground when high-frequency operational data are available, because they convert expected loads, door-opening effects and temperature trajectories into implementable setpoint schedules. DRL is better suited to highly dynamic facilities with nonlinear responses, but its performance depends on reward design, sensor placement, safe exploration and reliable enforcement of temperature constraints. Digital twins are most useful for complex refrigeration systems where thermodynamic mechanisms, equipment interactions and operating data must be combined, although they require sustained data flows, calibration and technical

expertise. In ageing or poorly maintained facilities, fault detection and predictive maintenance may provide the most immediate gains because soft faults, defrost failures and refrigerant-side abnormalities can increase energy use even when setpoints appear compliant.

Overall, smart energy management in cold stores is most credible when evidence is reported with paired quality, safety and efficiency indicators rather than energy metrics alone. Alongside electricity use, peak-demand reduction and system-efficiency metrics (e.g., COP), studies should disclose temperature-limit compliance (time above limit and excursion severity), disturbance performance during door events and defrost cycles and, where feasible, product quality or shelf life outcomes. Recent reviews of postharvest cold storage emphasise that future progress depends on integrating intelligent monitoring, data-driven control and digital-twin capabilities in ways that explicitly account for cost, produce quality and energy consumption (Hoffmann et al., 2025). From a mechanism-driven perspective, the central requirement is that energy-saving actions remain constrained by product-relevant thermal states, equipment operating states and food-safety limits, so that lower electricity use does not come at the cost of hidden quality loss or reduced operational resilience.

4.2. AI for reducing carbon footprint in cold-chain operations

Food cold chains are estimated to account for around 4% of total global greenhouse-gas emissions, but their carbon footprint cannot be assessed from energy use alone (UNEP and FAO, 2022). Emissions arise from refrigeration and transport energy, refrigerant leakage, operational disruptions and the embedded upstream emissions of food that is ultimately lost or wasted. From a sustainability perspective, cold chains therefore need to balance the resources required to maintain food quality and safety against the greenhouse-gas costs of cooling, transport and waste (Chen et al., 2022; Xu & Sun, 2025). This balance is critical because cradle-to-grave emissions from food loss and waste have been estimated at 9.3 Gt CO₂-eq globally, accounting for about half of total greenhouse-gas emissions from food systems (Zhu et al., 2023). AI-enabled carbon reduction should therefore be interpreted within life cycle assessment (LCA) boundaries that include both operational emissions and the avoided embedded emissions associated with reduced loss.

The relative importance of different mitigation pathways depends strongly on product type, system boundary and operating context. For cold-chain products in China, Dong and Miller (2021) showed that cold-chain energy use dominated many fruit and vegetable scenarios, whereas food loss and waste contributed more strongly to the footprint of meat and aquatic-product cold chains. This means that reducing refrigeration energy may be the priority in some product chains, while improving condition monitoring, shelf-life prediction and loss-prevention decisions may deliver larger carbon benefits in others. Luo et al. (2022) similarly showed that IoT-based monitoring can provide a carbon-reduction leverage effect, but that the magnitude of the benefit varies markedly across food categories, with greater potential for high-emission foods such as meat than for many fruits and vegetables. Recent dynamic shelf-life evidence further demonstrates how improved information can be translated into net climate benefits when embedded in decision rules: an IoT-DSL framework extended average shelf life by 8.1–13.8% for fruits, dairy and vegetables, avoided 17.32 ± 3.65 Mt yr⁻¹ of food waste and achieved a net reduction of 51.00 ± 10.38 Mt CO₂-eq yr⁻¹, despite additional sensor-related emissions (Wu, Zou, Liu, et al., 2025). These findings suggest that prediction, sensing and carbon mitigation should not be evaluated as separate outcomes, but as coupled effects on life cycle emissions per quality-safe delivered product.

Mechanism-driven AI can strengthen this coupling by linking time-temperature histories, quality-degradation kinetics, carbon accounting and operational constraints to actionable decisions. Coupled computational fluid dynamics and LCA can convert cold-chain interruptions into emission consequences by reconstructing dynamic temperature histories and estimating stage-specific refrigeration energy use; in a navel-orange

case, ambient exposure after pre-cooling increased global warming potential by up to 5.94%, while refrigerated transport contributed more than 85% of total GWP (Meng et al., 2026). The ENOUGH tool similarly integrates time-temperature profiles, refrigeration and transport emissions, packaging and food-quality kinetic models, enabling supply-chain scenarios to be compared in terms of both product quality and greenhouse-gas emissions (Leducq et al., 2026). Real-world transport carbon accounting further provides the operational baseline for such decisions, as shown by a bottom-up model developed from 128 long-haul refrigerated trips that covered driving, idling, refrigeration processes and refrigerant leakage (Lin et al., 2025). On this basis, AI can support carbon- and freshness-aware interventions, including route selection, dispatch adjustment and quality-constrained delivery scheduling. For example, target-oriented robust optimisation can minimise the risk of failing to meet freshness targets while accounting for cost and carbon emissions, showing that low-carbon routing is meaningful only when freshness and service constraints are explicitly included (Ding et al., 2025). Future studies should therefore report paired carbon and product-outcome metrics, including energy use, emission factors, refrigerant-related impacts, food loss or discard rates, RSL or DSL preservation, and final quality-safe delivery outcomes, rather than reporting prediction accuracy or energy saving as isolated indicators.

4.3. Waste minimisation and quality preservation

Food waste in cold chains is frequently a decision problem under uncertainty rather than a pure refrigeration failure: static date codes and incomplete visibility can lead to premature discard, delayed removal or misallocation, while waste may also increase under promotions, inventory imbalance and warehouse inefficiencies (Fatorachian & Shokri, 2025). Therefore, waste minimisation should be framed as quality-state-based preservation with actionable information, avoiding overly conservative operation that shifts losses from “spoilage” to “discard”. From a mechanism-driven perspective, the central task is to convert exposure histories and deterioration-relevant states into decision thresholds that indicate when a product should be sold, rerouted, inspected, reallocated or recovered before the remaining quality window closes.

DSL provides a practical bridge by updating RSL from real exposure using real-time data and/or indicators, such as time-temperature indicators (TTIs) or sensors, and communicating risk downstream (Wu, Zou, Chen, et al., 2025). This logic is consistent with IoT-enabled food-supply-chain architectures in which sensory input is processed into decision support and then translated into execution, including environmental adjustment, route changes or sales-strategy adaptation (Luo et al., 2022). However, evidence suggests that the benefit is conditional: information sharing reduces waste in well-controlled low-temperature settings, while in medium- and high-temperature environments DSL must be paired with timely interventions (Wu, Zou, Chen, et al., 2025). DSL should thus be embedded in an actuation layer, not treated as a standalone labelling upgrade. Its practical value depends on whether updated RSL estimates change downstream decisions, rather than simply improving the precision of shelf-life information.

Quality-aware actions can be grouped into three operational levers. First, allocation and dispatch can shift from first-in, first-out (FIFO) to first-expired, first-out (FEFO), and from batch-level to lot-level decisions, routing more fragile items to shorter paths or higher-turnover outlets while preserving longer-life items for longer routes. In this case, FEFO is not only an inventory rule but a quality-risk prioritisation strategy driven by updated RSL, uncertainty and product-condition estimates. Second, dynamic markdown and replenishment can use DSL-informed quality trajectories to accelerate sell-through before quality drops below acceptable thresholds. From a digital transformation perspective, implementing dynamic pricing for perishables requires more than algorithms: robust data capture, analytics integration and

continuous adaptation of pricing rules are needed to align operational practices, governance and store execution (Syed et al., 2024). Third, recovery and reverse flows can be redesigned as dynamic, data-driven networks that treat returns and unsold products not only as disposal problems but as opportunities for value recovery via reprocessing pathways. Integrated designs that combine real-time shelf-life data, ML and optimisation show how joint dynamic pricing and reverse logistics can reduce waste while maintaining profitability and social objectives (Geraeli & Roghanian, 2026). To lower deployment barriers, especially when single-use electronics become an environmental concern, emerging chipless, biodegradable, temperature-responsive tags can provide low-cost, disposable indication of temperature excursions without batteries or silicon, supporting scalable screening and exception handling (Bourelly et al., 2025).

Evaluation should jointly quantify quality, economics and sustainability: RSL preservation and time outside target ranges, alongside shrink and discard, margin, stockouts and feasibility (Syed et al., 2024;

Wu, Zou, Chen, et al., 2025). It should also distinguish genuine waste prevention from the displacement of losses across stages or into lower-value treatment routes, because food-loss and waste impacts depend on both where losses occur and how they are subsequently managed (Xue et al., 2024). Reducing food loss also avoids embedded emissions; DSL-based strategies can reduce both waste and associated greenhouse-gas emissions when coupled to appropriate operational decisions (Wu, Zou, Liu, et al., 2025). Overall, waste minimisation is a closed-loop decision process: sensing and state estimation yield decision-ready quality indicators, and downstream policies, including allocation, markdown, rerouting, inspection and recovery, convert them into verified reductions in waste and emissions, consistent with the execution architectures in Section 3. Accordingly, future studies should move beyond forecast accuracy alone and report intervention outcomes such as avoided discard, sell-through before quality thresholds, RSL or DSL preservation, redistribution or reprocessing rates, consumer-acceptable quality and evidence that losses have not merely

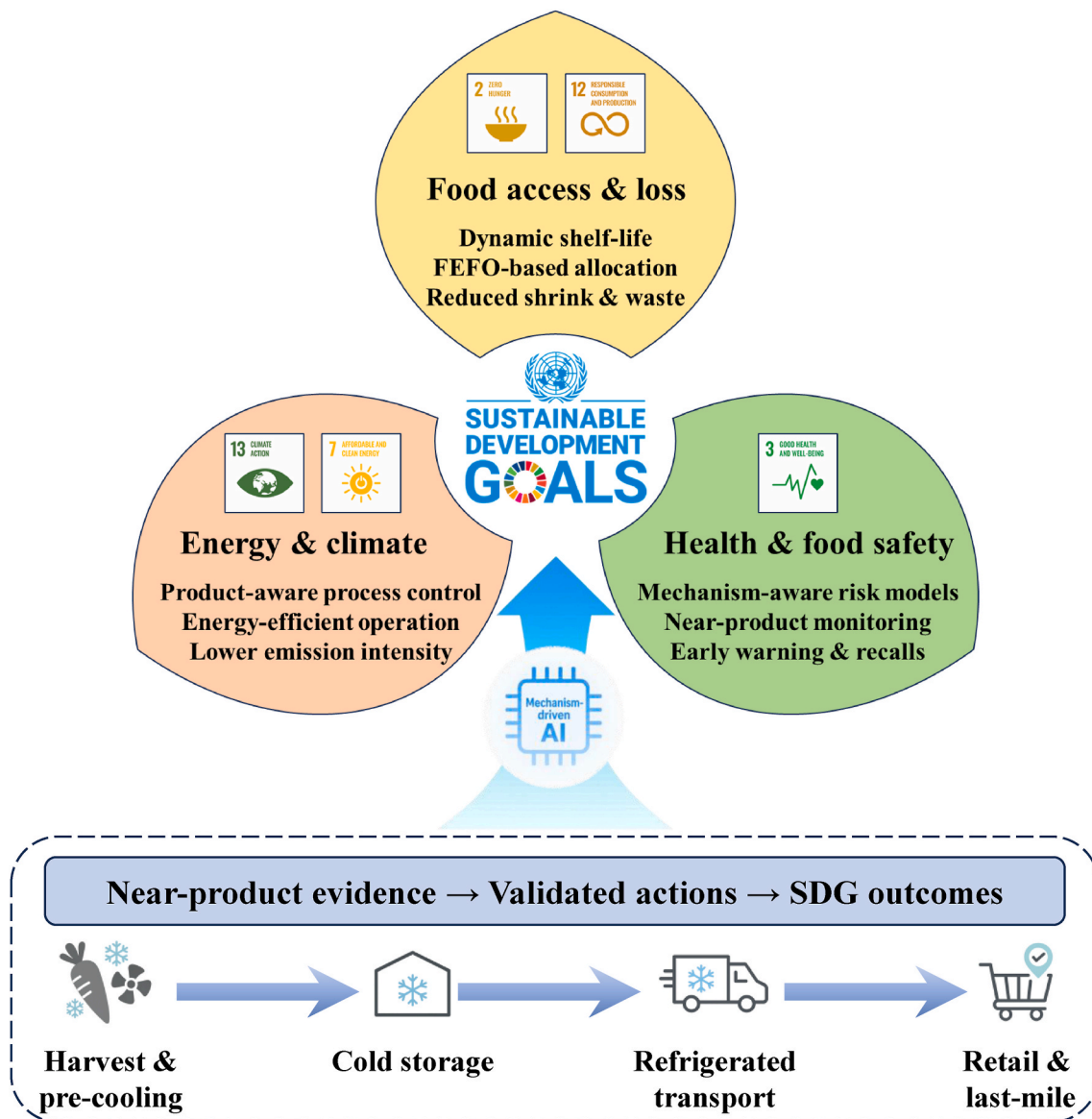


Fig. 4. Mechanism-driven AI pathways linking food cold-chain operations to Sustainable Development Goals. The framework summarises how mechanism-driven AI can support food access and loss reduction, energy and climate goals, and health and food safety across harvest and pre-cooling, cold storage, refrigerated transport, and retail and last-mile stages. The central pathway highlights that SDG-relevant outcomes depend on translating near-product evidence into validated actions, including dynamic shelf-life updating, FEFO-based allocation, product-aware process control, mechanism-aware risk modelling, early warning and targeted recalls.

been transferred to another supply-chain stage.

4.4. Contribution to Sustainable Development Goals

As summarised in Fig. 4, mechanism-driven AI can support several Sustainable Development Goals (SDGs), most directly SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-Being), SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action). These contributions are not created by AI deployment alone, but by the conversion of mechanism-linked evidence, including near-product exposure histories, deterioration-relevant states, equipment conditions and carbon accounting, into validated and traceable interventions. SDG alignment therefore depends on whether cold-chain systems reward quality protection, waste reduction, food-safety assurance and low-carbon operation as coupled outcomes rather than isolated technical indicators.

For food access and loss reduction (SDG 2.1 and SDG 12.3), the key pathway is to translate real exposure visibility into decision-ready quality indicators, such as remaining shelf life (RSL), dynamic shelf life (DSL) and quality-risk scores. These indicators can support first-expired, first-out (FEFO) allocation, rerouting, markdown, replenishment and redistribution before products fall below acceptable quality thresholds. In this sense, mechanism-driven AI contributes to food access not only by predicting deterioration, but by enabling edible products to be directed towards the most suitable market, processing or recovery pathway within their remaining quality window.

For energy and climate goals (SDG 7.3 and SDG 13), the contribution of AI should be assessed through quality-safe energy and carbon outcomes. Smart cold-store control, fleet optimisation, refrigerant management and low-carbon routing can reduce energy use and emission intensity, but only when cooling relaxation, route changes or

maintenance actions remain constrained by product condition and safety requirements. Life cycle-aware evaluation is therefore essential, because lower kWh consumption or shorter transport emissions may not represent genuine climate benefits if they increase food loss and the embedded emissions of wasted products.

For health and food safety (SDG 3), mechanism-driven AI strengthens early warning and risk prioritisation by linking near-product monitoring, time-temperature exposure, deterioration kinetics and equipment anomalies to safety-relevant decision thresholds. This can help identify vulnerable lots, trigger targeted inspection, support timely intervention and narrow recall scopes when incidents occur. At the same time, practical SDG contribution requires equity and deployability. High capital costs, proprietary platforms and complex compliance requirements may concentrate benefits in large integrated chains. Scalable impact therefore depends on affordable sensing, interoperable data standards, auditable decision rules and capacity-building, so that quality, safety and sustainability gains can be shared across actors, commodities and regions.

5. Challenges and future directions

Experience to date suggests that the limiting factors for AI in food cold chains lie less in model classes than in how data, mechanisms, constraints and decisions are connected in deployable systems. For mechanism-driven AI, the central challenge is not only to improve prediction or optimisation accuracy, but also to ensure that model outputs can be converted into reliable, traceable and quality-safe interventions under real cold-chain conditions. Key challenges include data quality and scarcity, data heterogeneity across products and operating contexts, explainability and auditability for decision support, transferability under domain shift and scalability under resource

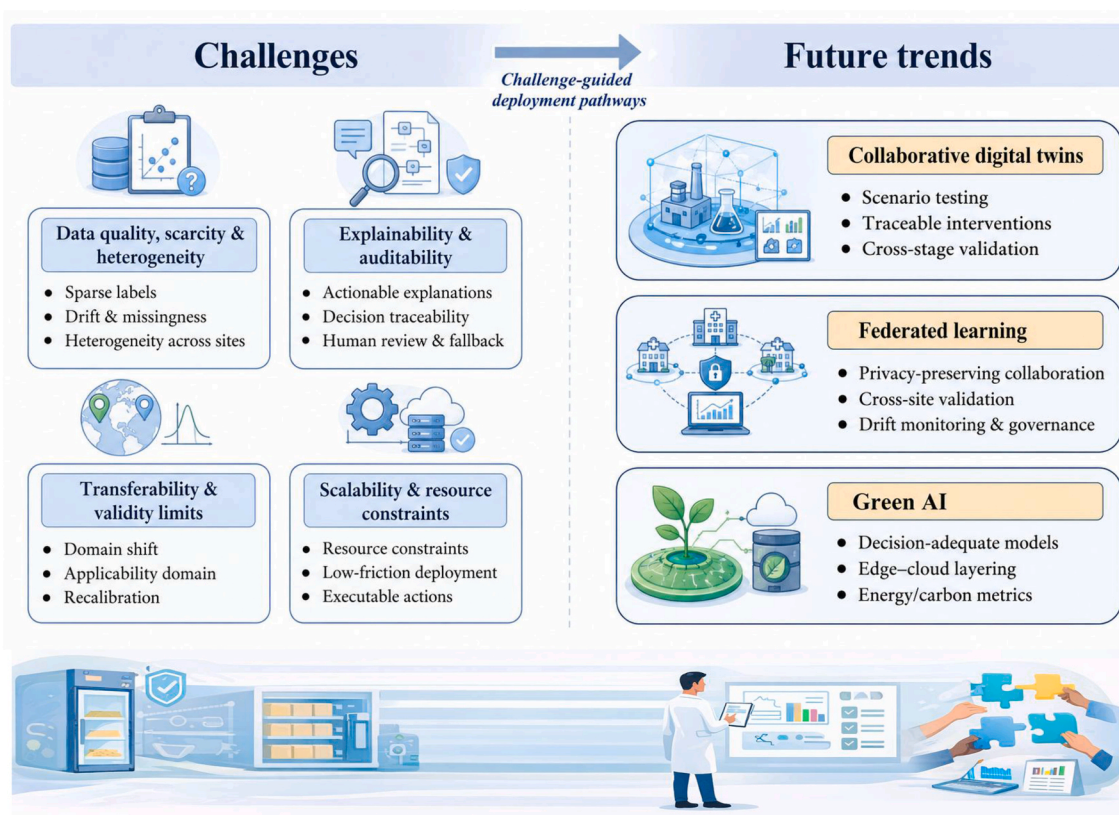


Fig. 5. Key challenges and challenge-guided future trends for mechanism-driven AI in food cold chains. Major challenges include data quality and heterogeneity, explainability and auditability, transferability and validity limits, and scalability under resource constraints. Corresponding future trends include collaborative digital twins, federated learning and Green AI, which together support more reliable, traceable and sustainable deployment of mechanism-driven AI across real cold-chain operations. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

constraints. These challenges should therefore be assessed by whether they improve decision reliability, intervention accountability and verifiable product, safety and sustainability outcomes, rather than by model sophistication alone. Fig. 5 summarises these challenge dimensions and highlights emerging trends that can help close the gap between promising prototypes and reliable, accountable deployment.

5.1. Data quality, scarcity and heterogeneity in cold-chain operations

In food cold chains, data heterogeneity refers not only to differences in data format, but also to differences in products, packaging, sensor placement, facility design, routes, seasons and decision contexts that make observations from one setting non-equivalent to those from another. Deployable AI in food cold chains is therefore limited less by model choice than by whether operational data can faithfully represent product exposure and the decision context that determines quality outcomes. In real operations, labels that truly reflect deterioration, such as shelf life, spoilage risk, rejection and claims, are often sparse, delayed and confounded by handling practices and downstream decisions, which makes “good retrospective accuracy” compatible with poor reliability under routine variability because models may learn site- or route-specific correlations rather than deterioration-relevant relationships (Göransson et al., 2018; Rashvand et al., 2025). At the same time, sensor streams are prone to drift, missingness and non-comparable sensor placement across facilities and vehicles, while data ownership is fragmented across actors, creating structural heterogeneity that amplifies domain shift when models are transferred across products, seasons and routes (Han et al., 2026).

Progress, therefore, depends on treating data as an engineered system with minimum requirements, rather than as an incidental by-product of operations. For mechanism-driven AI, these requirements are not merely technical data-management rules; they determine whether sensor streams can be linked to product exposure, deterioration mechanisms, state estimation and decision thresholds. At minimum, cold-chain datasets intended for prediction or control should include: (i) time synchronisation and stable device identity; (ii) interoperable metadata that records product and packaging, loading configuration, route and key operational events; (iii) routine calibration and drift detection; and (iv) auditable pipelines that preserve raw streams, document pre-processing, and report uncertainty and stratified performance by site, season and route (Han et al., 2026; Zhang, Liu, et al., 2025). Where direct labels are scarce, weak supervision and proxy targets can be used, but only when their validity is externally checked and their failure modes are explicitly characterised (Rashvand et al., 2025). Finally, shifting sensing closer to the product, for example, via packaging-integrated sensing coupled with AI, offers a path to more representative exposure measurement without requiring dense infrastructure, but still requires standardised data practices to be decision-ready (Jia et al., 2025).

5.2. Explainability, auditability, and responsible decision-making

In this review, explainability refers to whether AI outputs can be linked to mechanism-relevant causes and controllable operational factors, whereas auditability refers to whether the data, model, decision threshold and executed action can be reconstructed after deployment. Cold-chain decisions are safety and quality critical and are executed by multiple actors, so the goal is not interpretability “for its own sake” but interpretability that makes decisions actionable and accountable (Yu, Pu, et al., 2025). For mechanism-driven AI, an explanation should show how a predicted risk or recommended action relates to product exposure, deterioration kinetics, equipment state or operational disturbance, rather than only reporting feature importance. A model output becomes decision-ready only when it can be traced to plausible operational drivers, such as loading and airflow constraints, recurrent door-opening disturbances and unstable setpoints, and translated into

feasible interventions, such as extending pre-cooling, adjusting supervisory setpoints, triggering inspection, rerouting or reprioritising dispatch. When explanations fail to connect predictions to controllable levers, they may increase confidence without improving control, which is especially risky in high-stakes food contexts where predictive models must be embedded into safeguards and operational procedures (Balakrishnan et al., 2025). In agri-food quality assessment using high-dimensional sensing such as spectroscopy, the same gap appears in practice: users need interpretable evidence to trust, troubleshoot and redesign sensing and modelling pipelines, not merely to view importance scores (Ahmed et al., 2025).

Auditability is the second requirement. Decisions should be reconstructable with clear records of inputs, pre-processing choices, model versions, uncertainty, thresholds, human overrides and executed actions, enabling post-event review and continuous improvement. Responsible deployment also requires predefined decision authority, human-review points and fallback rules, especially when recommendations affect product release, rejection, rerouting, markdown or recall. Crucially, explainability must remain reliable under domain shift: cold-chain variability makes it easy for post-hoc narratives to look convincing while being unstable. Statistical perspectives on interpretable ML highlight that interpretation without stability checks and validation can mislead, motivating uncertainty estimation, shift monitoring and abstention or conservative fallbacks when conditions exceed the validated range (Allen et al., 2024). Thus, explainability and auditability should be evaluated by whether they improve intervention quality, traceability and accountability, not by whether they merely make model outputs appear more transparent.

5.3. Transferability and limits of validity

Transferability refers to the validated ability of a model or decision rule to remain reliable when applied beyond the conditions under which it was developed, while limits of validity define the conditions outside which its outputs should not be trusted without recalibration or review. Cold-chain AI must operate across products, packaging formats, facilities, climates and market practices, yet these factors jointly reshape both thermal exposure and deterioration pathways (Chen et al., 2026). As a result, a model can remain well calibrated in the setting it was developed for but become unreliable when moved to a different operating envelope, because the mapping from signals to quality outcomes changes with product physiology, packaging permeability, airflow and loading configurations and handling intensity (Iyer & Robb, 2025; Verreydt et al., 2024). This is a mechanism-level problem rather than only a statistical one: the same temperature history may imply different risks of respiration, moisture loss, microbial growth, texture degradation or shelf-life depletion across products and packages. In this setting, transferability is not a binary property of an algorithm; it is a property of the data–mechanism–decision pipeline and the conditions under which it was validated.

A deployment-ready approach, therefore, treats transfer as a managed process with explicit validity boundaries. The goal is controlled transfer, not unrestricted generalisation across all cold-chain settings. Practically, this means: (i) representing exposure and risk using descriptors anchored in mechanisms or stable operational quantities rather than convenience proxies; (ii) performing targeted recalibration using small but informative local data; (iii) defining an applicability domain and monitoring for domain shift; and (iv) reporting stratified performance by product, season, site and route instead of relying on a single aggregated metric. This evaluation discipline is central to “just evaluation” in food AI, where progress depends on revealing failure modes under realistic heterogeneity rather than masking them through averaging (Zhang, Liu, et al., 2025). When transfer remains uncertain, systems should default to conservative policies and human-in-the-loop verification to prevent confident but incorrect decisions. Such safeguards are especially important when model outputs trigger

interventions such as product release, rerouting, markdown, rejection or recall.

5.4. Scalability under resource constraints

Scalability refers to the ability of an AI-enabled cold-chain system to maintain reliable performance and accountable operation when deployed across many sites, products, routes and resource conditions. Scaling AI in food cold chains is therefore fundamentally a systems problem: many high-impact settings face constrained cooling capacity, intermittent power, limited connectivity and thin maintenance bandwidth, so solutions that rely on dense instrumentation, continuous cloud links, or frequent expert recalibration tend to fail operationally. In such settings, mechanism-driven AI must be translated into decision-adequate sensing, lightweight inference and robust fallback rules, rather than assuming ideal data streams and continuous connectivity. Modernisation analyses of legacy cold-storage infrastructure show that energy, economic and environmental outcomes are tightly coupled in real retrofits, meaning that deployment success depends on engineering feasibility and life cycle burden as much as on predictive performance (Singha et al., 2025). Likewise, technology constraints inside cold stores, non-uniform cooling, stratification, moisture loss, refrigerant choices and maintenance realities set hard boundaries on what “smart control” can reliably achieve without complementary upgrades (Hoffmann et al., 2025).

A deployment-ready notion of scalability, therefore, prioritises robustness, low operational friction and clear accountability. In practice, this means designing for graceful operation under outages and missing data; limiting required operator workload; and expressing model outputs as a small set of executable actions with explicit escalation paths when uncertainty is high, such as “hold and re-cool”, “inspect”, “reprioritise dispatch” or “reroute”. It also means evaluating scalability with deployment metrics rather than accuracy alone, including sensing and communication requirements, calibration and maintenance burden, failure handling, operator workload, user adoption and the persistence of benefits over seasons and sites, because cold-chain heterogeneity and harsh conditions routinely erode performance if monitoring and maintenance are treated as afterthoughts (Han et al., 2026). Scalable mechanism-driven AI should therefore be judged by whether it can preserve decision reliability and product protection under realistic resource constraints, not by whether it can deploy the most complex model architecture.

5.5. Future trends: toward secure, verifiable and sustainable AI deployment

Guided by the challenges above, future progress in mechanism-driven cold-chain AI should not be framed as the adoption of increasingly complex technologies, but as the development of deployment pathways that make sensing, prediction, intervention and documentation more reliable, collaborative and resource efficient. Digital twins, federated learning and Green AI are therefore considered here not as independent trends, but as complementary responses to the challenges of validation, data fragmentation, transferability and resource-constrained deployment.

5.5.1. From single-link digital twins to whole-chain collaborative twins

Digital twins are no longer only a speculative future technology in cold-chain systems; equipment-, facility- and shipment-level twins have already been used to support monitoring, state estimation, what-if testing and control assistance. The future trend is therefore not the simple adoption of digital twins, but their evolution from single-link or asset-level twins towards whole-chain collaborative twins that connect products, equipment, logistics events, energy and carbon information, and intervention records across cold stores, transport, retail and last-mile operations. This direction is consistent with recent food-supply-

chain digital-twin implementation frameworks, which emphasise that wider adoption requires not only technical modelling capability, but also structured pre-adoption assessment, adoption planning, post-adoption evaluation and stakeholder coordination (Huang et al., 2024). In this form, digital twins can shift cold-chain management from monitoring air temperatures to shipment-specific and chain-level decision support by continuously coupling sensor streams with mechanistic or hybrid models that estimate product-relevant states and quality risks. A twin can be used as a scenario-testing engine to answer “what-if” questions under realistic constraints, such as delays, setpoint deviations and door-opening events, so that alerts and interventions are triggered based on predicted quality trajectories rather than static thresholds (Defraeye et al., 2021). In citrus supply chains, a physics-based twin translates in-transit histories into actionable metrics, including moisture loss, microbial spoilage risk, chilling injury, pest mortality and remaining life, making trade-offs explicit and reframing control objectives from temperature compliance to operating within acceptable quality–marketability ranges (Shrivastava et al., 2022).

The next step is to embed such twins within collaborative cyber–physical service layers so that scenario testing, intervention rehearsal, execution and post-event review can be linked across actors and stages. Integration of digital twins with AI and shelf-life kinetics has demonstrated that modest sensing can support low prediction error for in-box temperature and day-level shelf-life estimation, and simulations indicate measurable reductions in food loss across multiple fruit types, illustrating a concrete pathway from prediction to intervention design (Zou et al., 2025). To make such interventions auditable at scale, twin outputs must be connected to service platforms that support real-time monitoring, anomaly detection, stakeholder interaction and decision logging, so that inputs, model versions, uncertainty, decisions and actions are recorded and reviewable (Wu et al., 2023). Whole-chain collaborative twins should therefore be evaluated not only by model fidelity, but also by whether they improve cross-stage validation, intervention traceability, accountability and verifiable outcomes such as quality retention, reduced food loss, energy flexibility and carbon-aware operation.

5.5.2. Federated learning for privacy-preserving and cross-site validation

Food cold chains are governed by multiple actors, so the data needed for robust prediction and control are naturally fragmented across growers and packers, cold stores, carriers, retailers, and platform or testing providers. This fragmentation is not only a privacy or ownership problem; it also limits external validation, domain-shift detection and learning from rare but high-impact deviations that may be distributed across many sites. Federated learning (FL) offers a practical collaboration mechanism in this setting: partners can train shared models while keeping raw data local, which lowers legal and commercial barriers to cross-site learning and helps models experience a broader range of operating conditions than any single organisation can provide (Dembani et al., 2025; Han et al., 2026). Cross-silo federated learning is particularly aligned with supply-chain realities because it enables learning across institutions without requiring direct data exchange, providing an explicit mechanism for “data sharing without sharing data” (Durrant et al., 2022).

However, FL should not be treated as an automatic solution to cold-chain heterogeneity. If participating sites differ in products, packaging, sensor placement, metadata quality, operating rules and decision thresholds, federated updates may still produce poorly calibrated or unevenly reliable models. For mechanism-driven cold-chain AI, federated collaboration therefore needs more than privacy-preserving optimisation; it requires shared definitions of product-relevant states, exposure descriptors, uncertainty reporting, evaluation strata and model-update rules. For deployment, heterogeneity and privacy risk must be managed explicitly through stratified evaluation and drift monitoring, together with update protection and auditable governance, so stakeholders know when federated models are dependable and when conservative fallbacks or human verification are required (Dembani

et al., 2025; Zhang, Liu, et al., 2025). In this role, FL becomes not only a training strategy, but also a governance mechanism for cross-site validation, controlled transfer and accountable learning across fragmented cold-chain networks.

5.5.3. Green AI for sustainable deployment in cold-chain systems

Green AI reframes progress in cold-chain intelligence as an efficiency–performance trade-off rather than accuracy alone. Rather than focusing only on the carbon footprint of individual models, Green AI provides a broader framework for measuring and reducing the resource costs of AI across model design, training, inference, communication, hardware and deployment workflows (Peykani et al., 2026-a; Verdecchia et al., 2023). This is particularly important for mechanism-driven AI because high-fidelity models, digital twins and federated learning can improve validation and collaboration, but they may also increase computation, communication and maintenance burdens. Green AI should therefore be understood not as a rejection of advanced modelling, but as a requirement that model complexity, update frequency and deployment architecture remain proportional to decision value and operational risk. Because cold-chain decisions increasingly rely on frequent, always-on inference, such as monitoring, alerting, route or setpoint adjustments, the operational footprint of deployed models can accumulate over long service lifetimes. Accordingly, model selection should consider not only predictive quality and decision effectiveness, but also energy use, carbon intensity, communication load, retraining frequency, hardware requirements and maintenance burden.

In practice, “greener” deployment means prioritising decision-adequate models that minimise computation, communication and maintenance burdens while preserving safety and quality constraints. A practical architecture is to use high-fidelity models and digital twins mainly for offline scenario testing, calibration, stress testing and policy design, while deploying compressed models, simplified decision rules or threshold-based fallbacks at the edge for routine monitoring and real-time execution. For edge- and IoT-dominated cold-chain settings, energy-aware design extends beyond model compression to system-level choices such as communication-efficient collaboration, energy-conscious client participation, specialised or low-power hardware, and privacy-preserving FL with lower communication and retraining overhead (Alahmari & Alghamdi, 2025). Framed this way, Green AI becomes a measurable deployment discipline: models are compared not only by predictive utility but also by the energy, computational and carbon intensity required to deliver that utility reliably in real operations (Peykani et al., 2026-a). Future studies should therefore report, where feasible, inference frequency, communication volume, retraining requirements, edge-device energy demand, cloud-computing use, hardware assumptions and maintenance burden, together with product protection, safety compliance and sustainability outcomes.

6. Conclusions

This review highlights that the most actionable opportunities in cold-chain AI come from shifting decision-making from air-temperature compliance to product-state evidence. From this perspective, mechanism-driven AI should not be understood as a new algorithmic label, but as a deployment logic that links near-product sensing, deterioration mechanisms, state estimation, constrained optimisation and governed action. Such evidence is generated by coupling near-product sensing with mechanism-informed models, including kinetics-informed, physics-aware and product-state-based formulations. It delivers value only when predictions are embedded in constrained, auditable control and governance loops and translated into closed-loop execution. The central contribution of this review is therefore to reposition cold-chain AI from model-centred prediction towards mechanism-linked, product-centred and governance-ready intervention.

However, translating these advances into reliable and deployable

impact still faces several persistent gaps. Auditable data pipelines and standardised product-relevant exposure metrics remain uneven; external-by-design validation across seasons, routes, facilities, products and subgroups is often limited; uncertainty, applicability domains and controlled transfer across contexts are still insufficiently reported; and end-to-end governance, including traceable decision logging, accountability, human review and integration with food-safety management systems, continues to lag behind model development. These shortcomings indicate that the main barrier is not simply the lack of more complex models, but the incomplete connection between sensing quality, mechanism-relevant state estimation, validated decision thresholds and executable interventions.

Looking forward, mechanism-driven AI has strong application prospects in smart and sustainable cold chains. At the product level, it can support DSL updating, RSL estimation, FEFO allocation, targeted inspection, redistribution and more precise recall decisions. At the process and logistics levels, it can enable quality- and safety-constrained control of refrigeration, freezing, thawing, storage and transport, while linking energy saving and carbon reduction to verified product outcomes. Future cold chains should therefore evolve into outcome-based systems in which collaborative digital twins support scenario testing and cross-stage validation, FL enables privacy-preserving cross-site learning and controlled transfer, and Green AI promotes decision-adequate models for low-burden edge deployment. Performance should be evaluated not only by prediction accuracy, cost reduction or on-time delivery, but also by verifiable quality retention, RSL preservation, time outside target conditions, food loss, intervention effectiveness, energy use, carbon outcomes and traceable accountability. Under these conditions, mechanism-driven AI can become a practical foundation for cold-chain governance that is measurable, trustworthy and scalable to operators of different sizes.

Contributions

First author Writing - original draft, Formal analysis, Investigation.

Second author Validation, Funding acquisition, Resources.

Last author Supervision, Funding acquisition, Resources, Writing - review & editing.

AI use declaration

Artificial intelligence-based tools were used solely for language-related assistance, including improvements in grammar, spelling, and overall clarity of author-written text. No AI system was used to generate scientific content, perform data extraction, conduct analyses, or influence the interpretation of results. All methodological decisions, data synthesis, and conclusions were developed, verified, and approved by the authors, who take full responsibility for the accuracy and integrity of the manuscript.

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Data availability

The authors are unable or have chosen not to specify which data has been used.

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